CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of B. Tech. Examination

December 2012

ME 102 Fundamentals of Mechanical Engineering

Date: 06.12.2012, Thursday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) What is energy? What are the types of energy? What is energy conversion? **[03]**

(b) Explain advantages and disadvantages of liquid fuels over solid fuels. **[04]**

Q - 2 (a) Define specific heat. Write the value of index n for isothermal and constant pressure process in PV^n=C law. **[02]**

(b) What is isothermal process? Derive an expression for the work done during the isothermal process. **[04]**

(c) 0.5 m^3 air fuel mixture at 1 bar and 100 °C is compressed as per law $PV^n = C$ until its pressure becomes 15 bar and temperate becomes 450 °C. Then it is ignited at constant volume and its pressure becomes double of the pressure at the end of compression. Calculate (i) T_{max} (ii) change in internal energy (iii) heat transferred. **[08]**

Take R = 0.298 kJ/kg K, γ = 1.4, n=1.3

OR

Q - 2 (a) Define dryness fraction and wetness fraction of steam. What do you understand by the term priming? **[03]**

(b) Why steam is superheated? Give advantages and disadvantages of superheated steam. **[03]**

(c) How much heat is required to convert 4 kg of water at 20°C into steam at 8 bar and 200°C. Take Cps= 2.1 KJ/KgK and Cpw=4.187 KJ/KgK. **[08]**

Q - 3 (a) What is the difference between boiler and steam generator? **[02]**

(b) Explain with neat sketch the working of Cochran boiler. **[06]**

(c) A steam generator evaporates 18000 kg/hr of steam at 10 bar pressure and steam is 97% dry. Feed water temperature is 40°C. Coal is fired at the rate of 2050 Kg/hr. Calorific value of coal is 28000 KJ/Kg. Calculate thermal efficiency and equivalent evaporation of the boiler. **[06]**

OR

Q - 3 (a) What is the difference between mountings and accessories. [02]

(b) Explain any two. [06]

1. Dead weight safety valve
2. Water level indicator.
3. Economiser.

(c) A boiler produces 1000 kg of steam/hr at pressure of 10 bar abs and dryness fraction 0.95 from feed water at 50°C. It is ascertained that 10% of coal is unburnt at the grate. If the efficiency of the boiler is 75%. Find the efficiency of the boiler and grate combined. The calorific value of coal is 31000 KJ/kg. Take Cpw=4.187 KJ/kgK [06]

SECTION – II

Q - 4 (a) Why priming is required in centrifugal pump? [02]

(b) Explain with neat sketch the working of centrifugal pump. [05]

Q - 5 (a) Derive equation for efficiency of Otto cycle with its p-v diagram. [06]

(b) During testing of single cylinder two stroke oil engine following data were obtained. Brake torque = 640 N.m., cylinder diameter =21 cm, speed =350 rpm, stroke =28cm, mean effective pressure =5.6 bar, oil consumption =8.16 kg/hr. calorific value of oil =42705 KJ/kg. Determine (a) mechanical efficiency (b) indicated thermal efficiency (c) brake thermal efficiency (d) brake specific fuel consumption. [08]

OR

Q - 5 (a) Explain the construction of vane type compressor with neat sketch. [06]

(b) A single stage air compressor is required to compress 90 m3/min of air from 1 bar pressure and 27°C temperature to 10 bar. Find the temperature at the end of compression, work done, power required and heat rejected during isothermal, adiabatic and polytropic process ($PV^{1.25}$=C). Neglect clearance effect. [08]

Q - 6 (a) Explain in detail of vapour compression refrigeration system? [06]

(b) Explain Oldham's coupling with neat sketch. [04]

(c) Show the comparison between flat belt and V- belt drive. [04]

OR

Q - 6 (a) Write short note on window air conditioner. [06]

(b) Distinguish between coupling and cluth. [04]

(c) Give classification of gears and explain any one of them with neat sketch. [04]

******